

Announcements

- Project 3 is due **Tuesday, February 27**, 11:59pm PT
- HW4 out later this week; due **Friday, March 1, 11:59pm PT**
- Midterm: **Tuesday, March 5, 7pm PT** (more info on website)

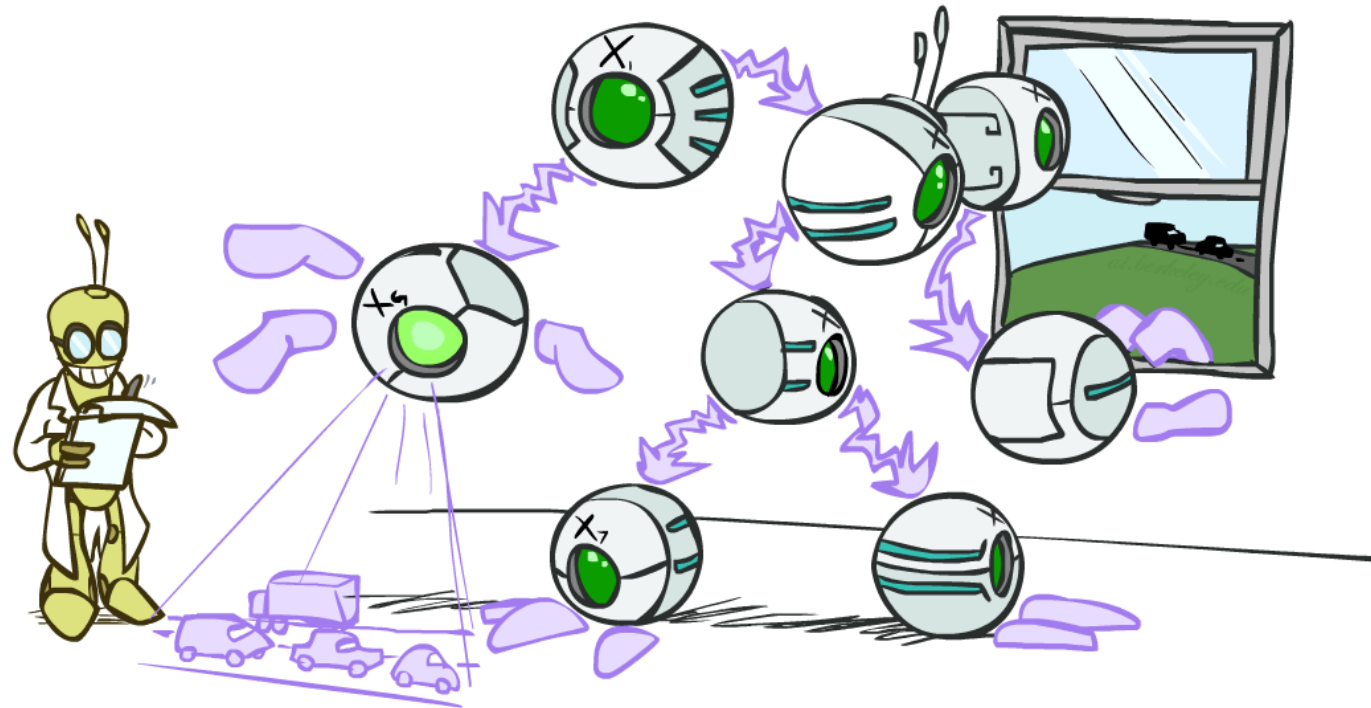


Pre-scan attendance
QR code now!

(Password appears later)

CS 188: Artificial Intelligence

Bayes Nets: Exact Inference



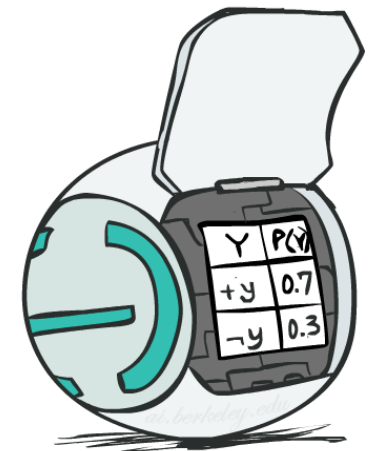
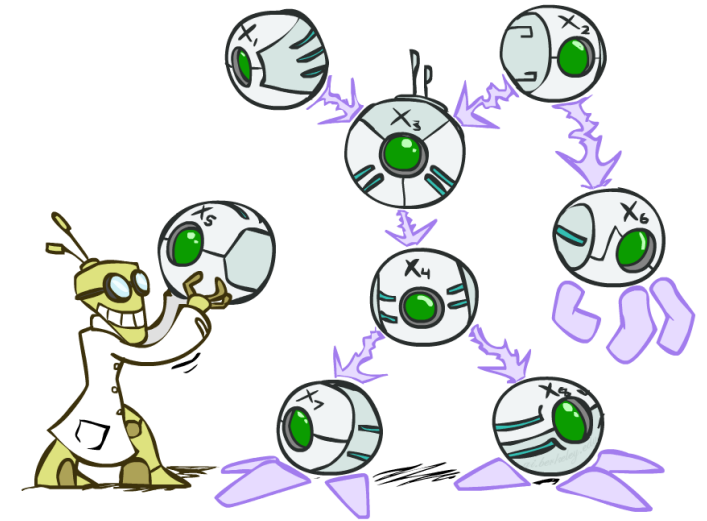
Bayes Net Representation

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

- Bayes nets implicitly encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



Inference

- Inference: calculating some useful quantity from a joint probability distribution

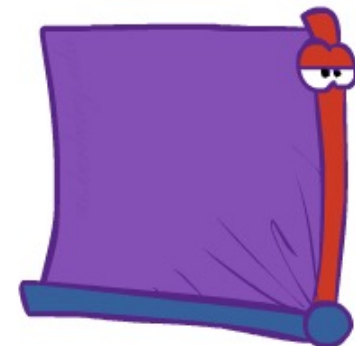
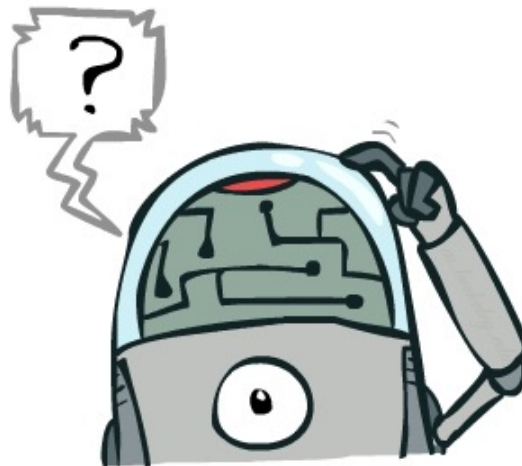
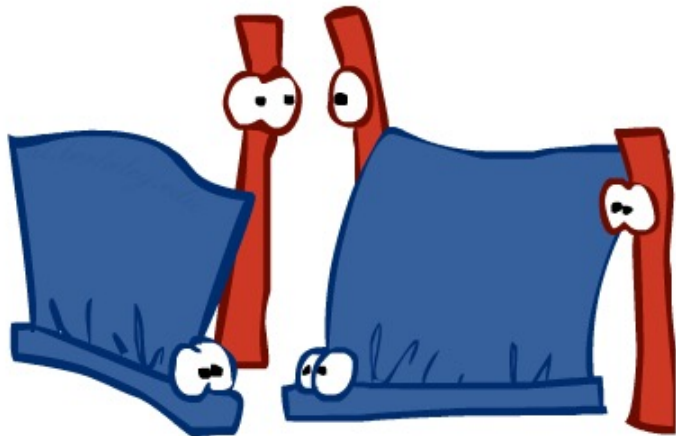
- Examples:

- Posterior probability

$$P(Q|E_1 = e_1, \dots, E_k = e_k)$$

- Most likely explanation:

$$\operatorname{argmax}_q P(Q = q|E_1 = e_1 \dots)$$



Inference by Enumeration

- General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $$\left. \begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array} \right\} \begin{array}{l} X_1, X_2, \dots X_n \\ \text{All variables} \end{array}$$

- We want:

** Works fine with multiple query variables, too*

$$P(Q|e_1 \dots e_k)$$

- Step 1: Select the entries consistent with the evidence

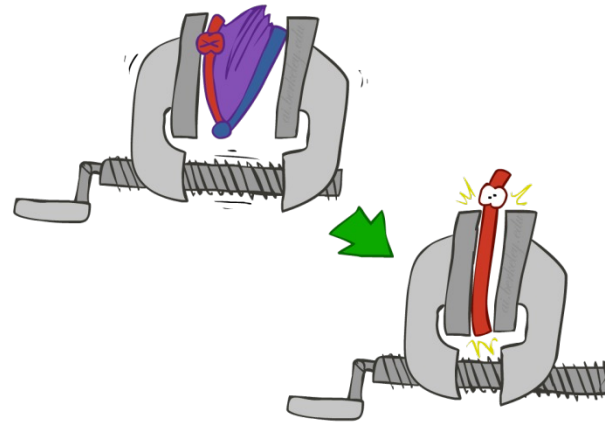


x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

2

0.15

- Step 2: Sum out H to get joint of Query and evidence



- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots X_n}$$

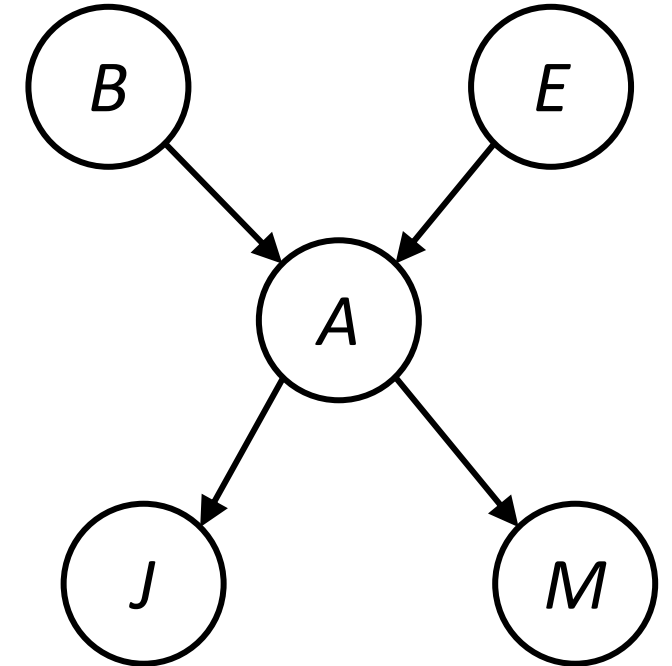
Inference by Enumeration in Bayes Nets

- Given unlimited time, inference in BNs is easy
- Reminder of inference by enumeration by example:

$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

$$= \sum_{e,a} P(B, e, a, +j, +m)$$

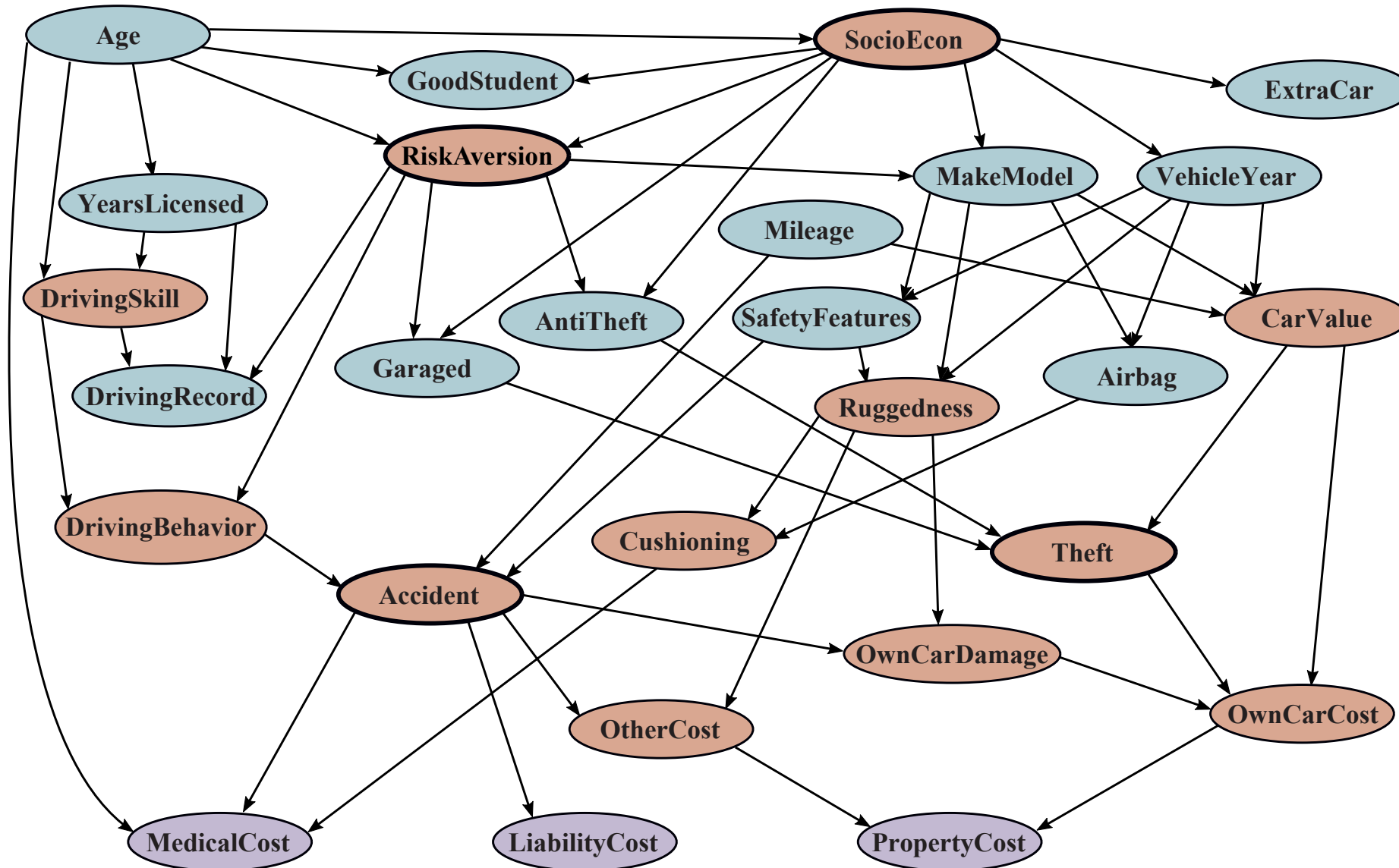
$$= \sum_{e,a} P(B)P(e)P(a|B, e)P(+j|a)P(+m|a)$$



$$= \begin{matrix} P(B)P(+e) & P(+a|B, +e) & P(+j|+a)P(+m|+a) & + & P(B)P(+e) & P(-a|B, +e) & P(+j|-a)P(+m|-a) \\ P(B)P(-e) & P(+a|B, -e) & P(+j|+a)P(+m|+a) & + & P(B)P(-e) & P(-a|B, -e) & P(+j|-a)P(+m|-a) \end{matrix}$$

Lots of repeated subexpressions!

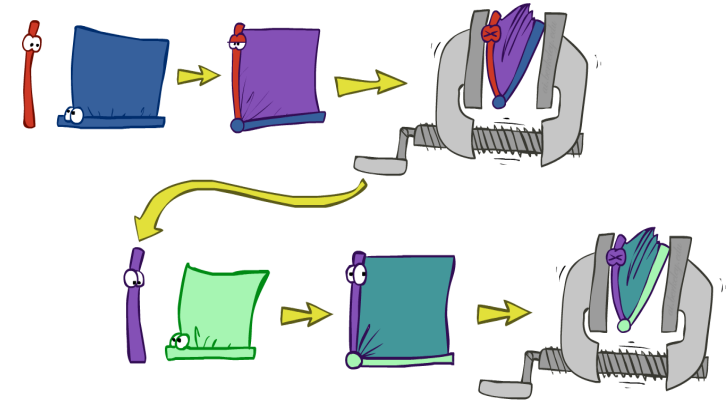
Inference by Enumeration?



Variable elimination: The basic ideas

- Consider: $uw y + uw z + ux y + ux z + vw y + vw z + vxy + vxz$
 - 16 multiplies, 7 adds
- Rewrite as: $(u+v)(w+x)(y+z)$
 - 2 multiplies, 3 adds
- Move summations inwards as far as possible
 - $P(B \mid j, m) = \alpha \sum_{e,a} P(B) P(e) P(a \mid B, e) P(j \mid a) P(m \mid a)$
 - $= \alpha P(B) \sum_e P(e) \sum_a P(a \mid B, e) P(j \mid a) P(m \mid a)$
- Do the calculation from the inside out
 - i.e., sum over a first, then sum over e
 - Note: $P(a \mid B, e)$ isn't a single number, it's a table!

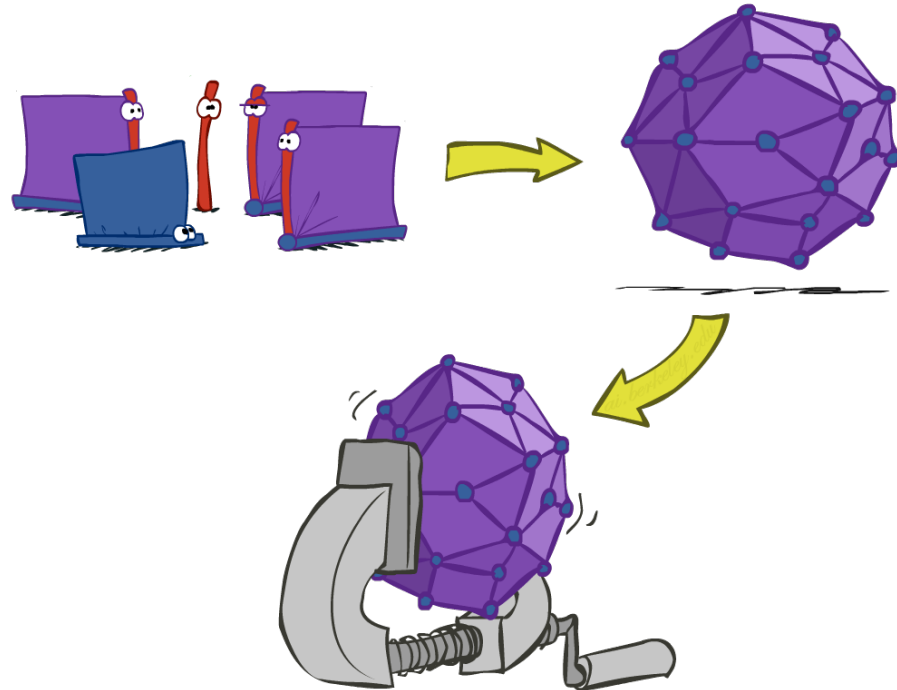
B should not depend on the latter events



Inference by Enumeration vs. Variable Elimination

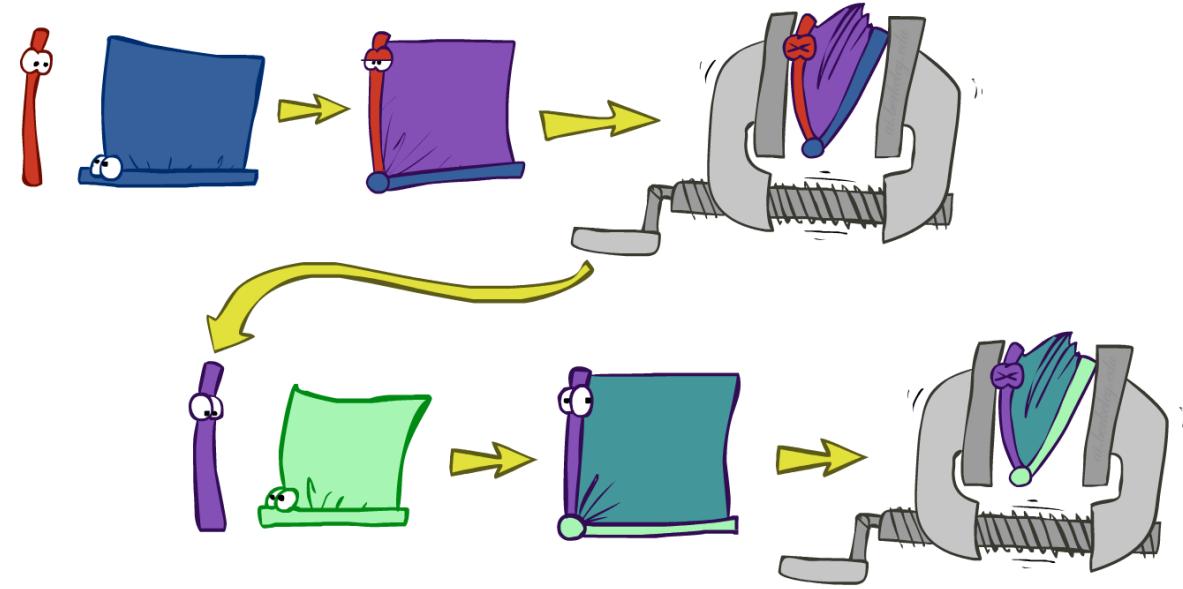
- Why is inference by enumeration so slow?

- You join up the whole joint distribution before you sum out the hidden variables



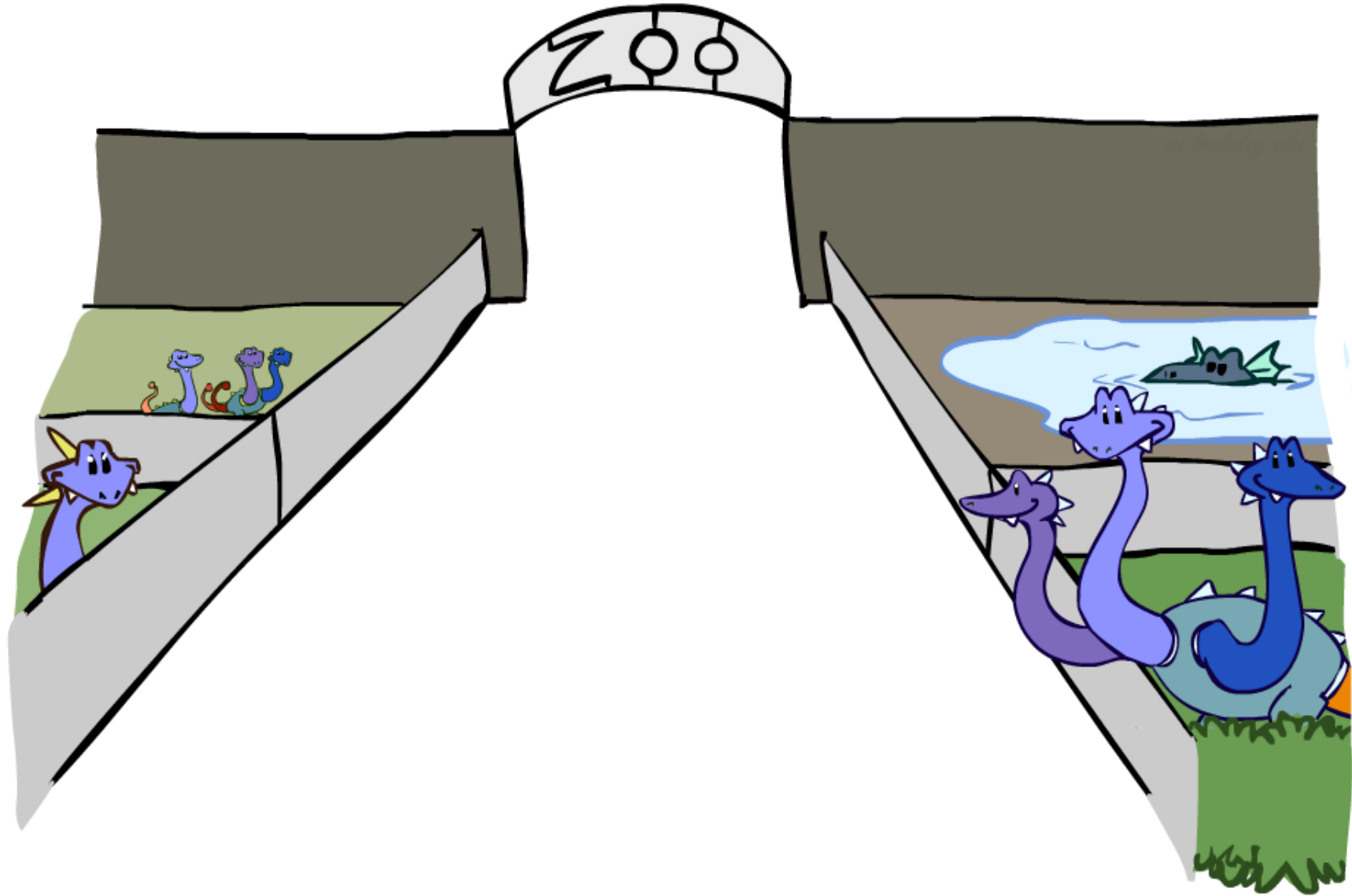
- Idea: interleave joining and marginalizing!

- Called “Variable Elimination”
- Still NP-hard, but usually much faster than inference by enumeration



- First we'll need some new notation: factors

Factor Zoo



Factor Zoo I

- Joint distribution: $P(X,Y)$

- Entries $P(x,y)$ for all x, y
- Sums to 1

$P(T, W)$

	W	
T	sun	rain
hot	0.4	0.1
cold	0.2	0.3

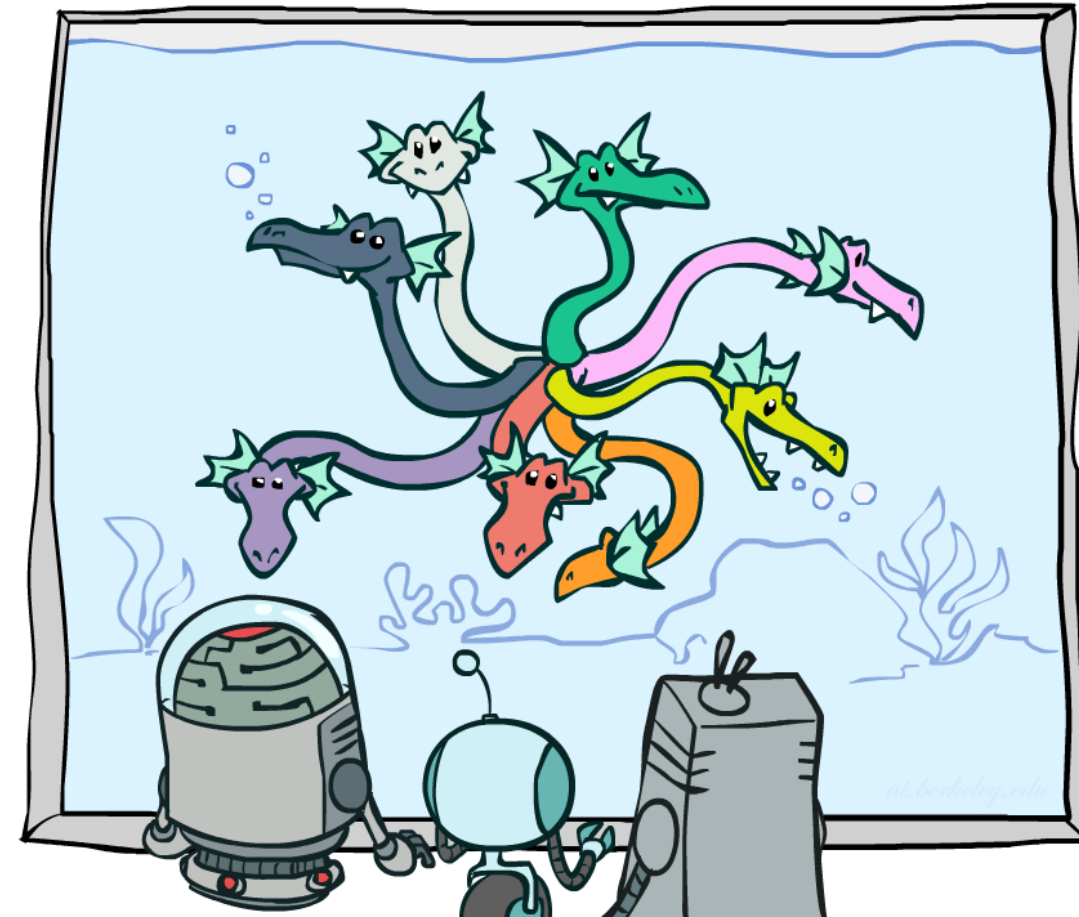
- Selected joint: $P(x,Y)$

- A slice of the joint distribution
- Entries $P(x,y)$ for fixed x , all y
- Sums to $P(x)$

$P(cold, W)$

	W	
T	sun	rain
cold	0.2	0.3

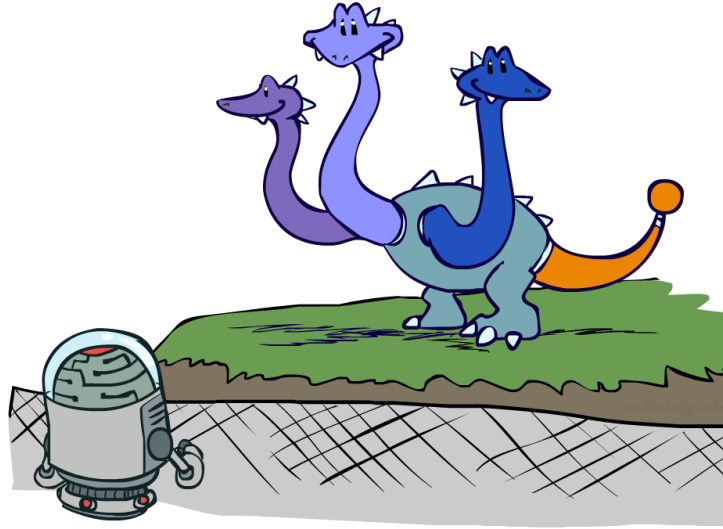
- Number of capitals = dimensionality of the table



Factor Zoo II

- Single conditional: $P(Y \mid x)$

- Entries $P(y \mid x)$ for fixed x , all
- Sums to 1

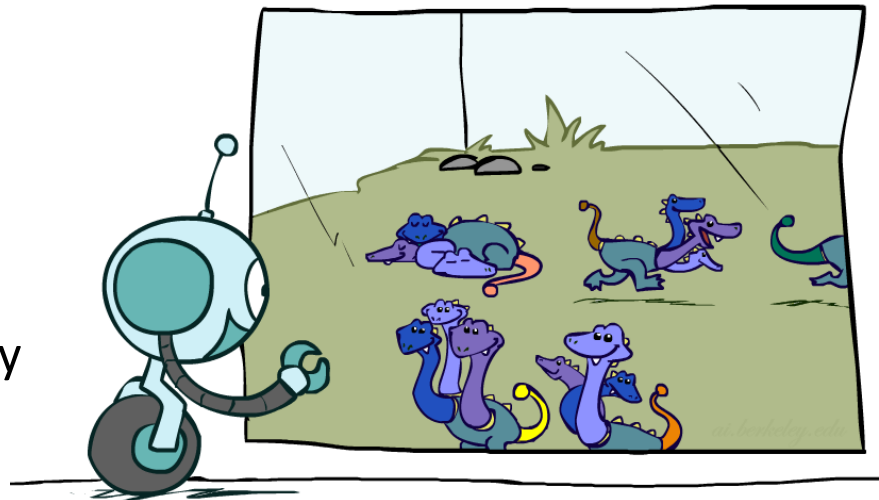


$$P(W \mid \text{cold})$$

	W	
T	sun	rain
cold	0.4	0.6

- Family of conditionals:
 $P(Y \mid X)$

- Multiple conditionals
- Entries $P(y \mid x)$ for all x, y
- Sums to $|X|$



$$P(W \mid T)$$

	W	
T	sun	rain
hot	0.8	0.2
cold	0.4	0.6

} $P(W \mid \text{hot})$
} $P(W \mid \text{cold})$

Factor Zoo III

- Specified family: $P(y \mid X)$

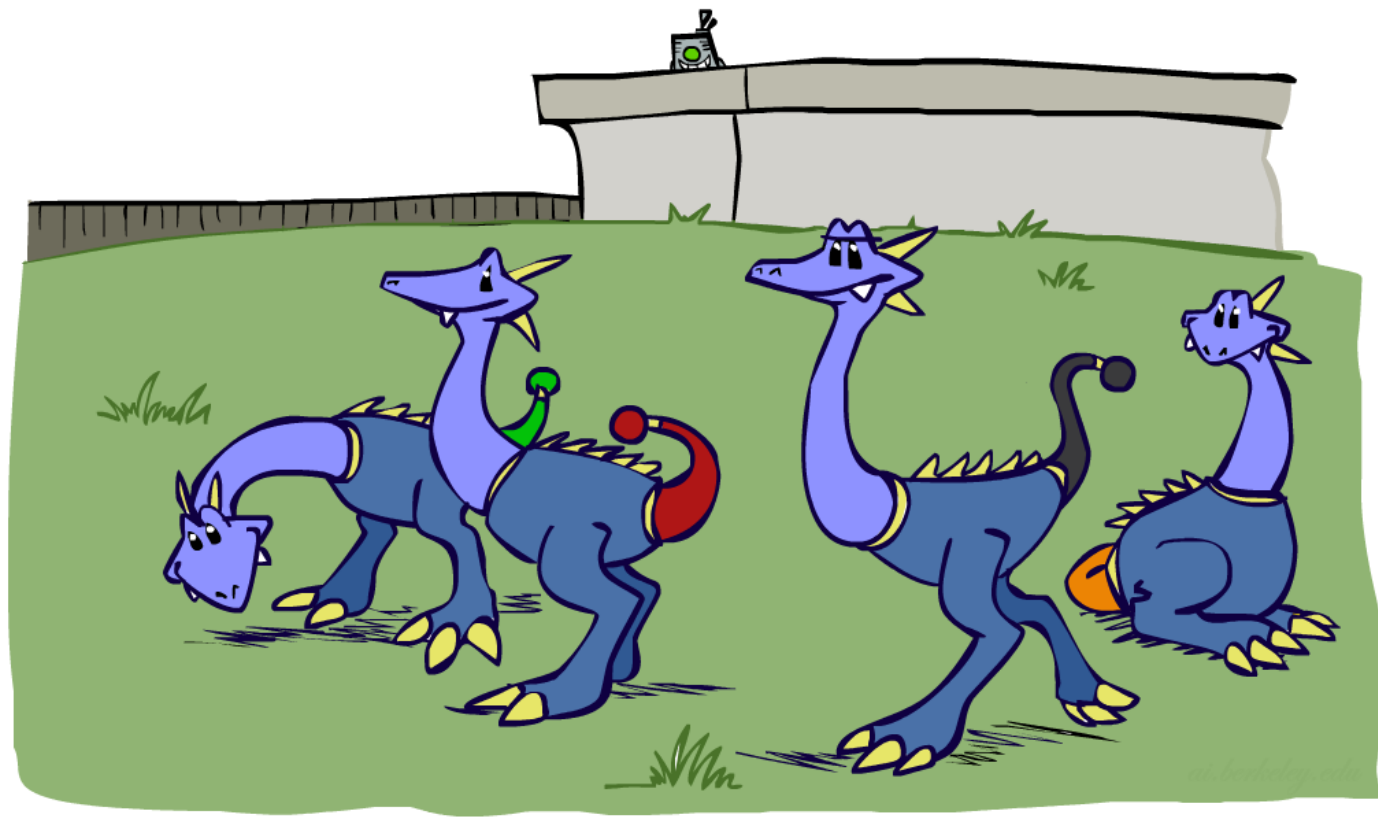
- Entries $P(y \mid x)$ for fixed y , but for all x
- Sums to ... who knows!

$$P(\text{rain} \mid T)$$

	W
T	rain
hot	0.2
cold	0.6

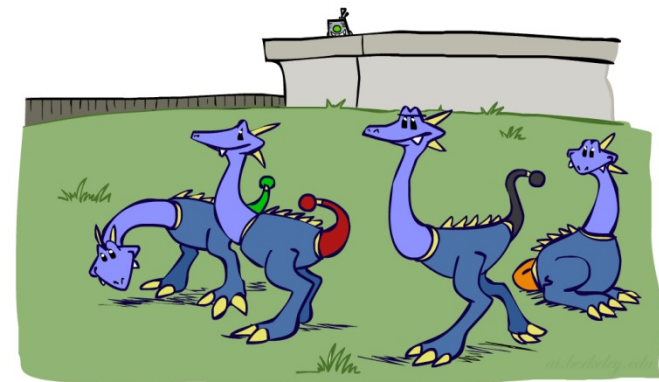
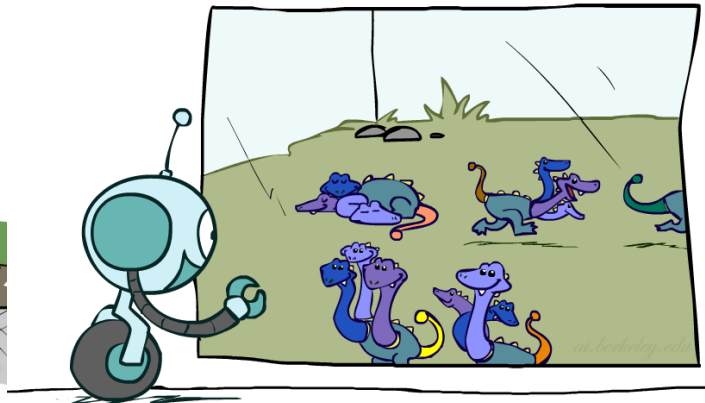
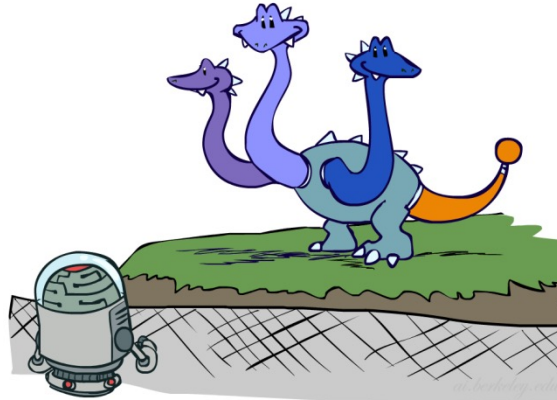
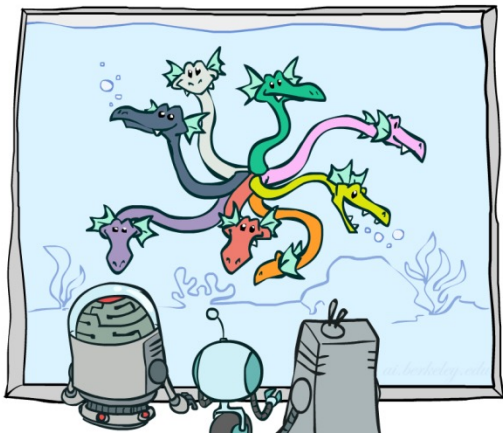
$$P(\text{rain} \mid \text{hot})$$

$$P(\text{rain} \mid \text{cold})$$

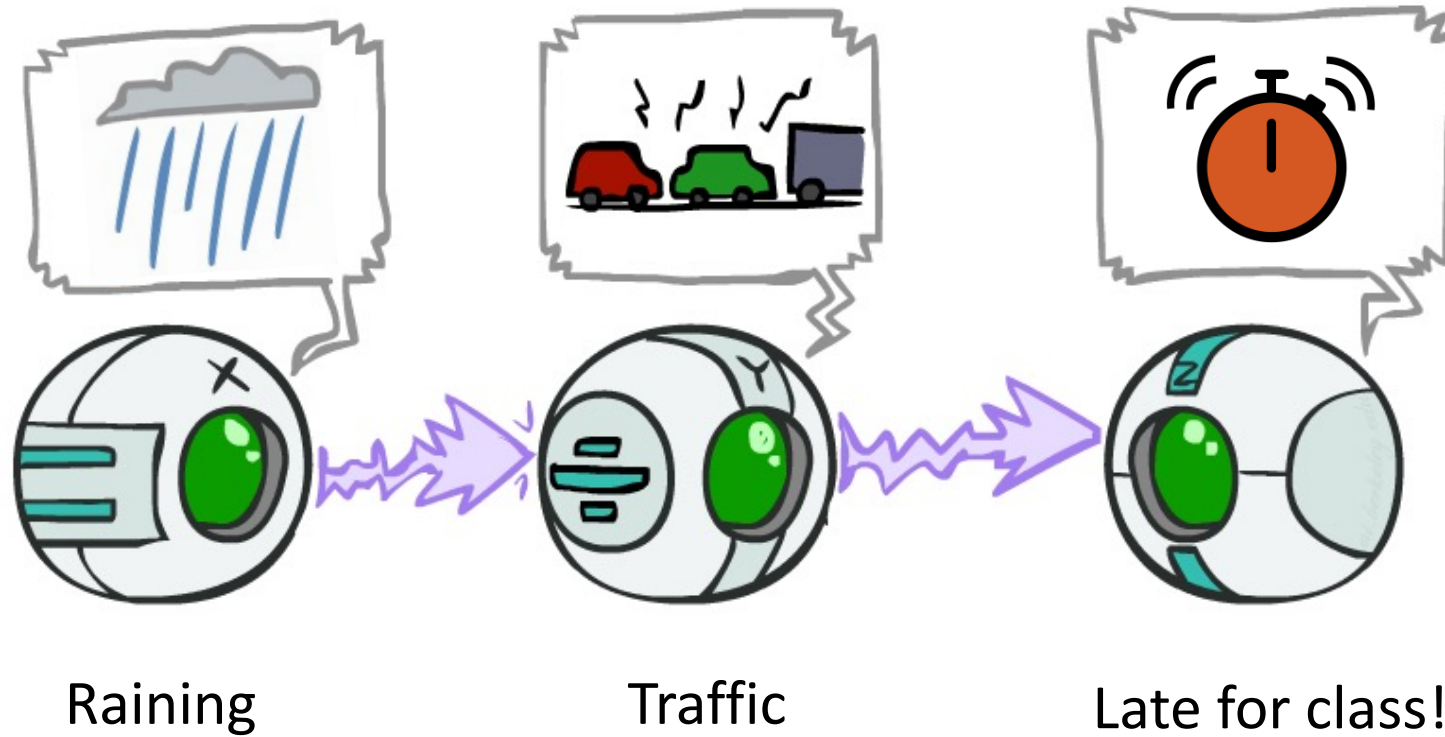


Factor Zoo Summary

- In general, when we write $P(Y_1 \dots Y_N \mid X_1 \dots X_M)$
 - It is a “factor,” a multi-dimensional array
 - Its values are $P(y_1 \dots y_N \mid x_1 \dots x_M)$
 - Any assigned (=lower-case) X or Y is a dimension missing (selected) from the array
 - Sometimes we'll write $P(A,b \mid c,D)$ as $f_i(A,b,c,D)$ —just another name for the same table.



Traffic Domain



Example: Traffic Domain

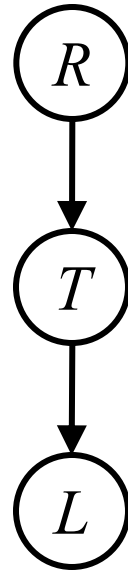
■ Random Variables

- R: Raining
- T: Traffic
- L: Late for class!

$$P(L) = ?$$

$$= \sum_{r,t} P(r, t, L)$$

$$= \sum_{r,t} P(r)P(t|r)P(L|t)$$



$$P(R)$$

+r	0.1
-r	0.9

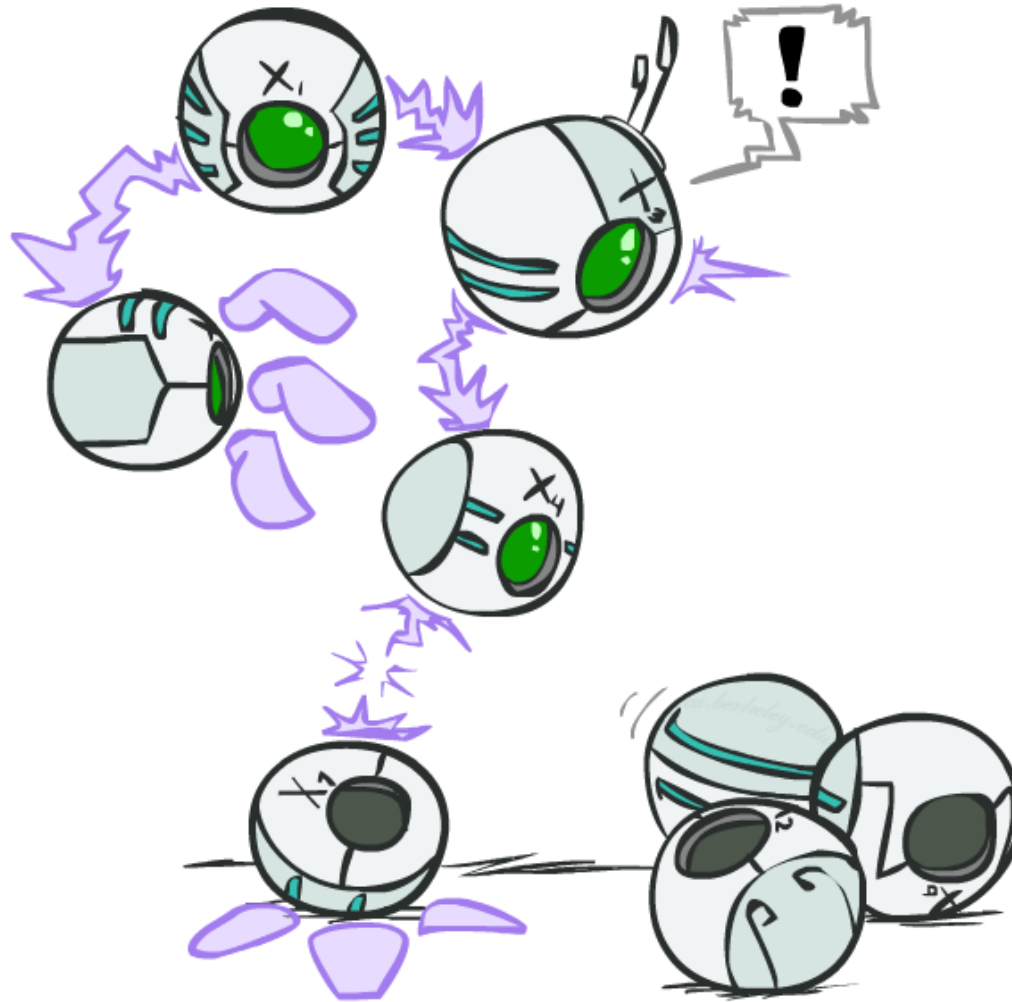
$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Variable Elimination (VE)



Inference by Enumeration: Procedural Outline

- Track objects called **factors**
- Initial factors are local CPTs (one per node)

$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- Any known values are selected
 - E.g. if we know $L = +\ell$, the initial factors are

$$P(R)$$

+r	0.1
-r	0.9

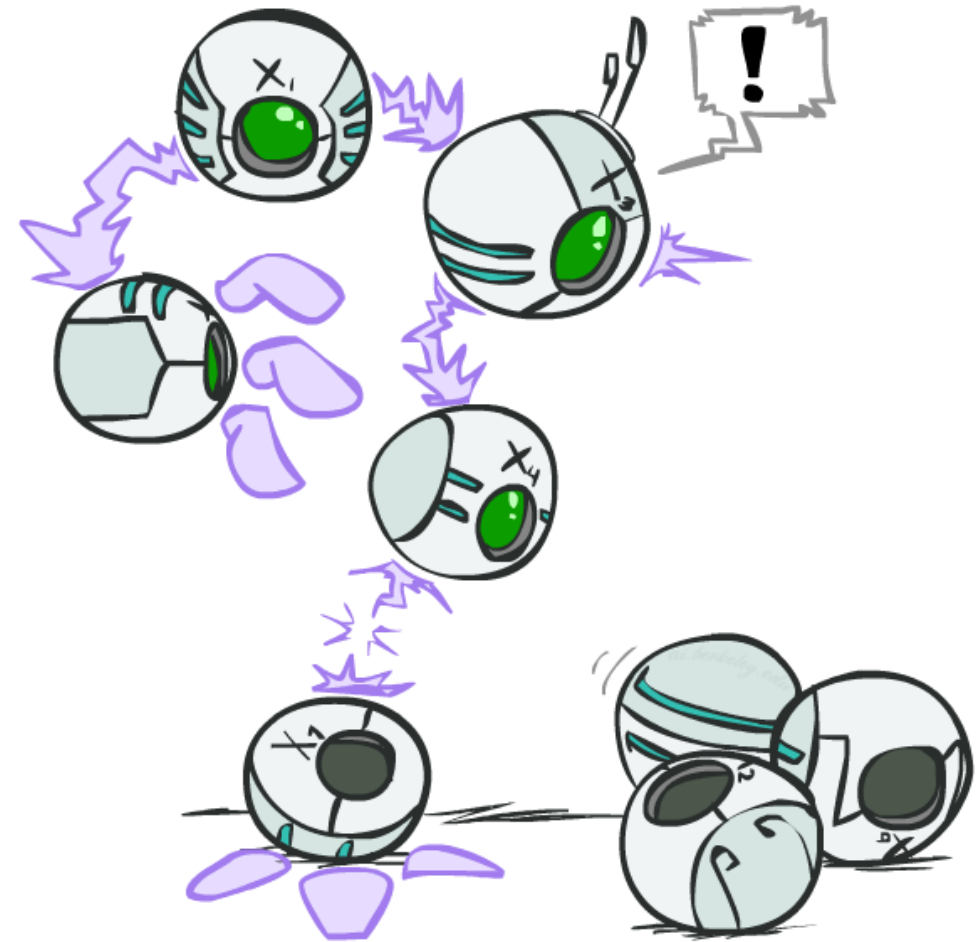
$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(+\ell|T)$$

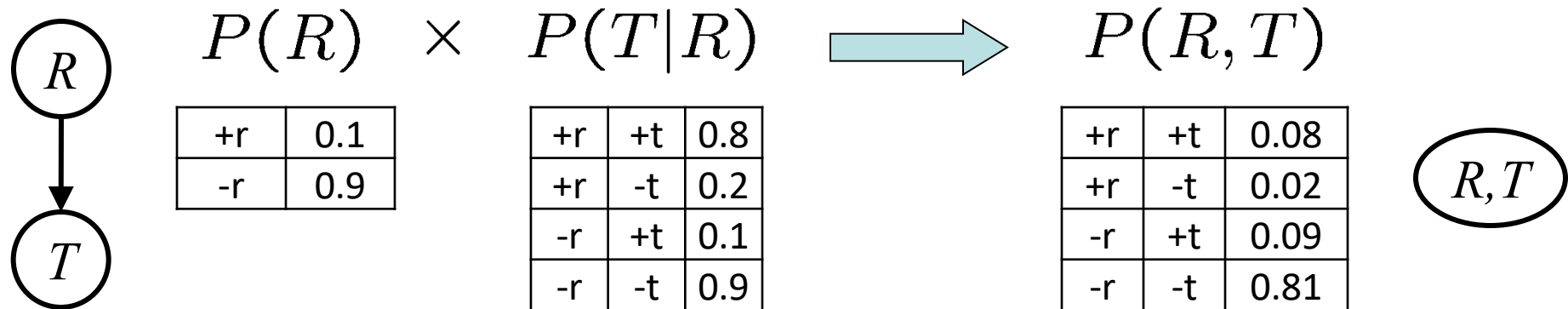
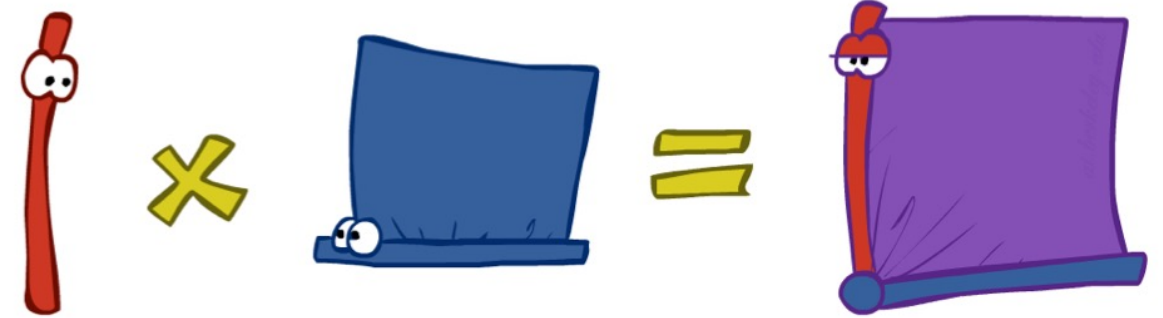
+t	+l	0.3
-t	+l	0.1

- Procedure: Join all factors, eliminate all hidden variables, normalize



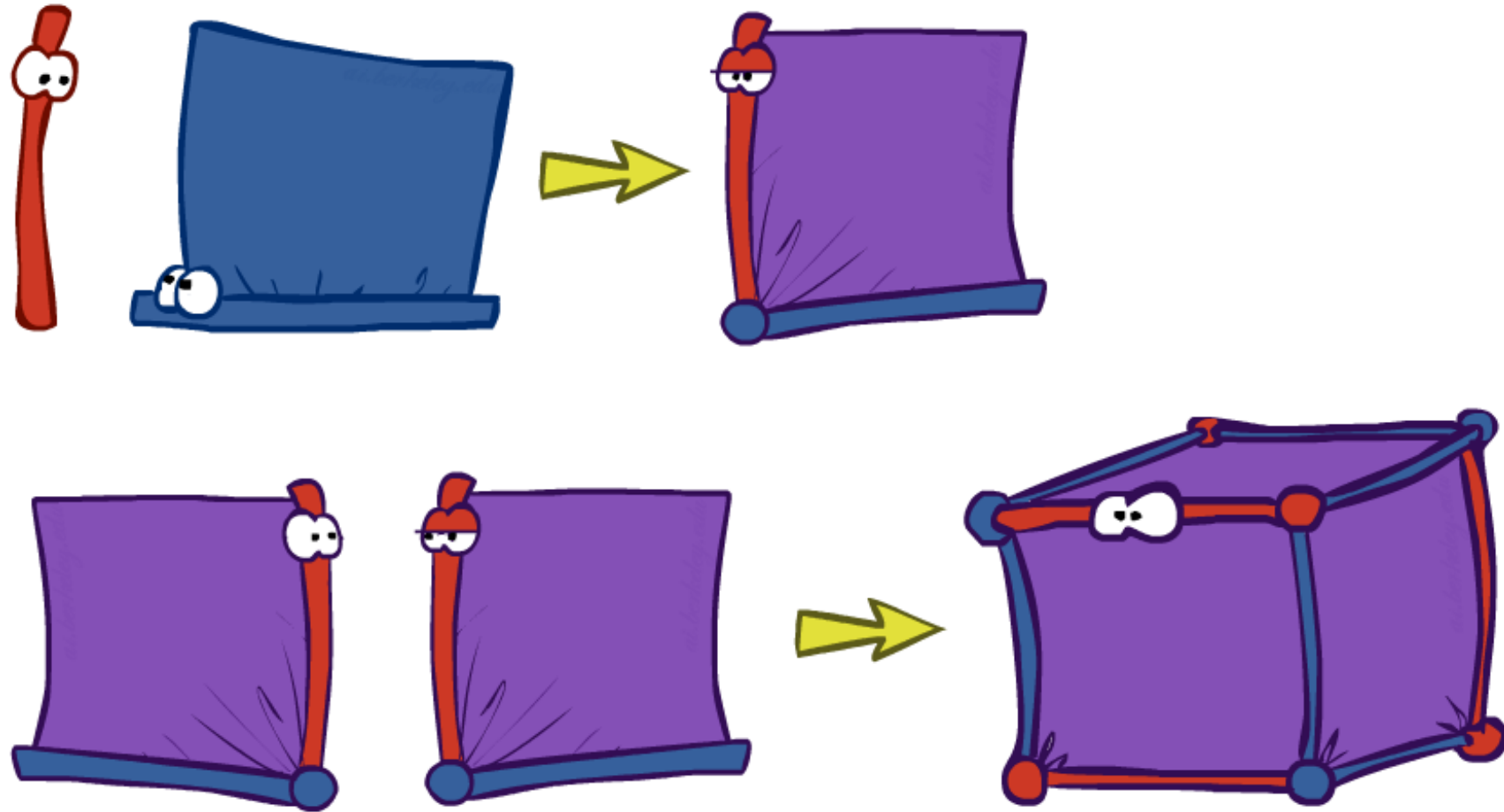
Operation 1: Join Factors

- First basic operation: **joining factors**
- Combining factors:
 - Just like a database join**
 - Get all factors over the joining variable
 - Build a new factor over the union of the variables involved
- Example: Join on R

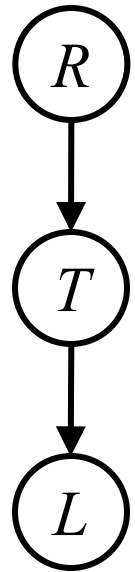


- Computation for each entry: pointwise products $\forall r, t : P(r, t) = P(r) \cdot P(t|r)$

Example: Multiple Joins



Example: Multiple Joins



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Join R

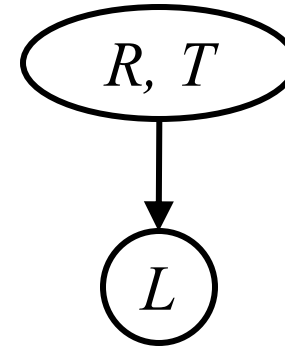


$P(R, T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9



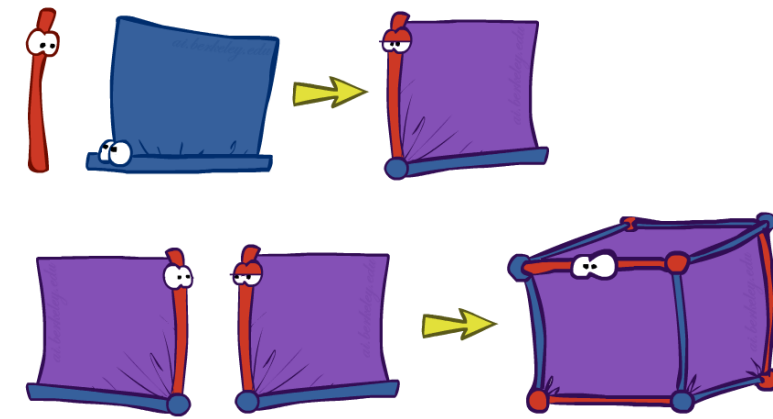
Join T



R, T, L

$P(R, T, L)$

+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729



Operation 2: Eliminate

- Second basic operation: **marginalization**
- Take a factor and sum out a variable
 - Shrinks a factor to a smaller one
 - A **projection** operation
- Example:

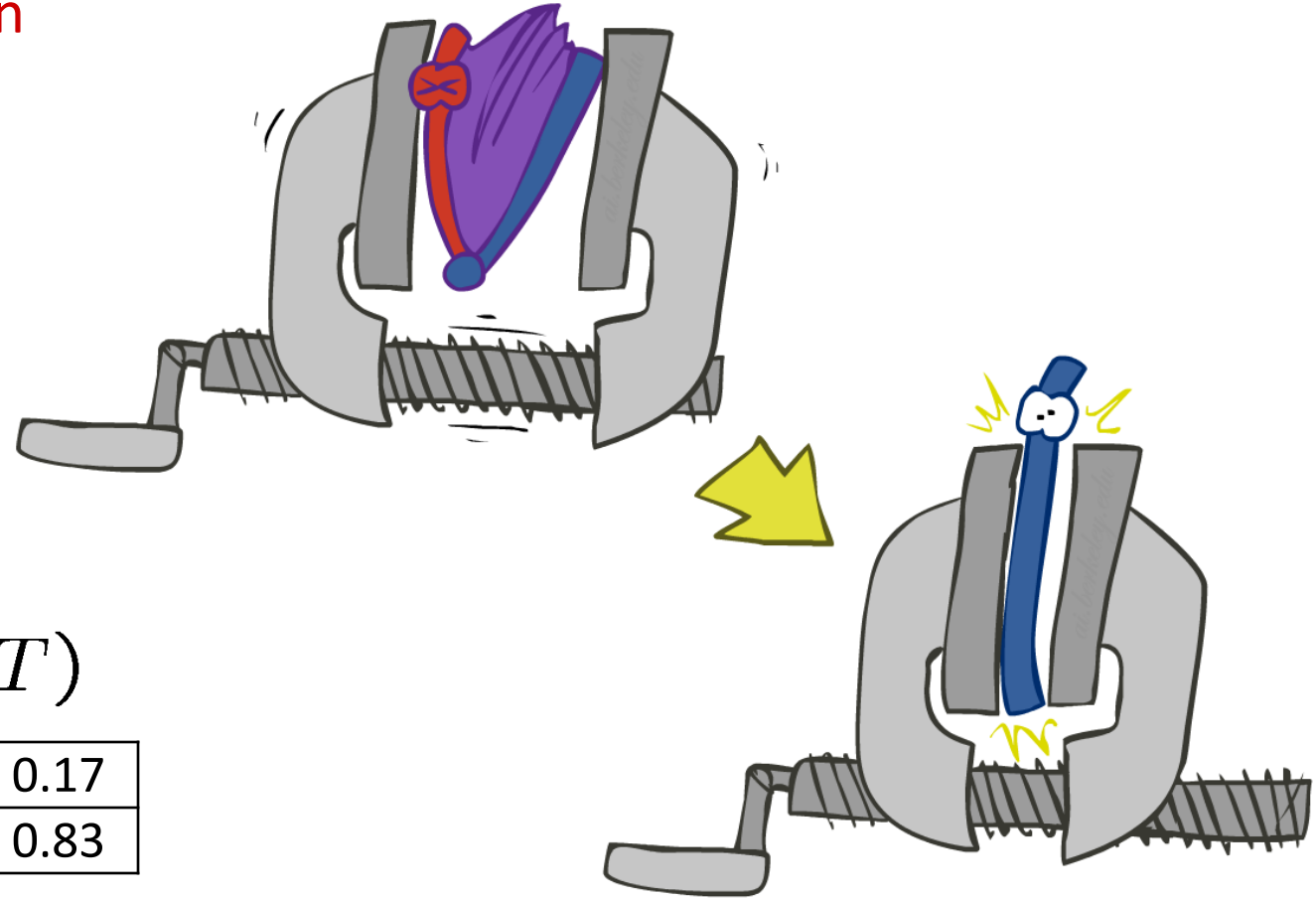
$$P(R, T)$$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

sum R


$$P(T)$$

+t	0.17
-t	0.83



Multiple Elimination

$P(R, T, L)$

R, T, L			
+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Sum
out R

→

$P(T, L)$

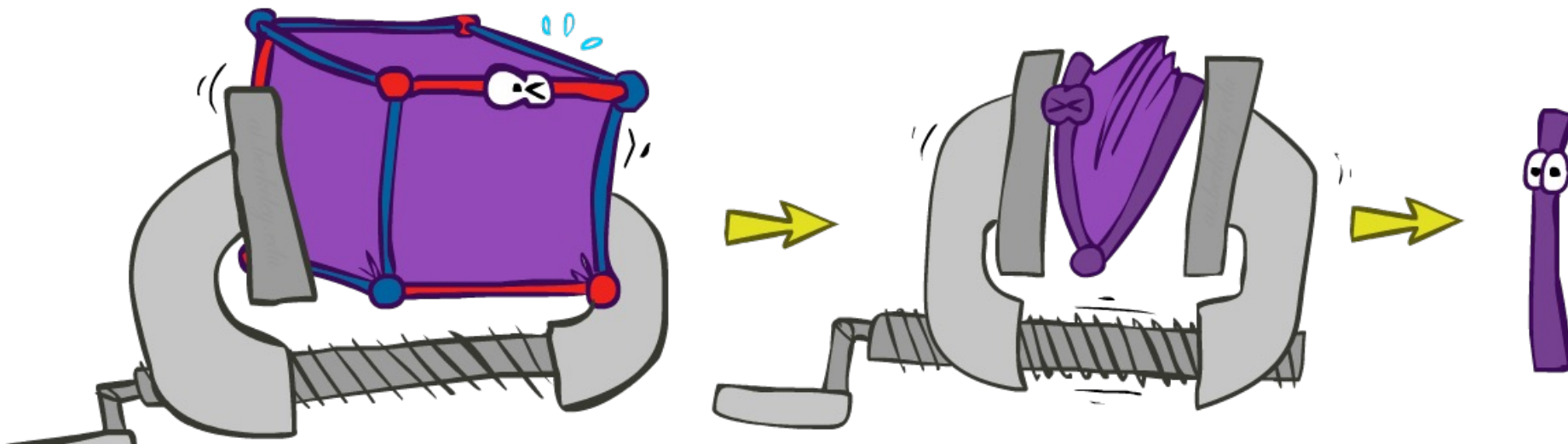
+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum
out T

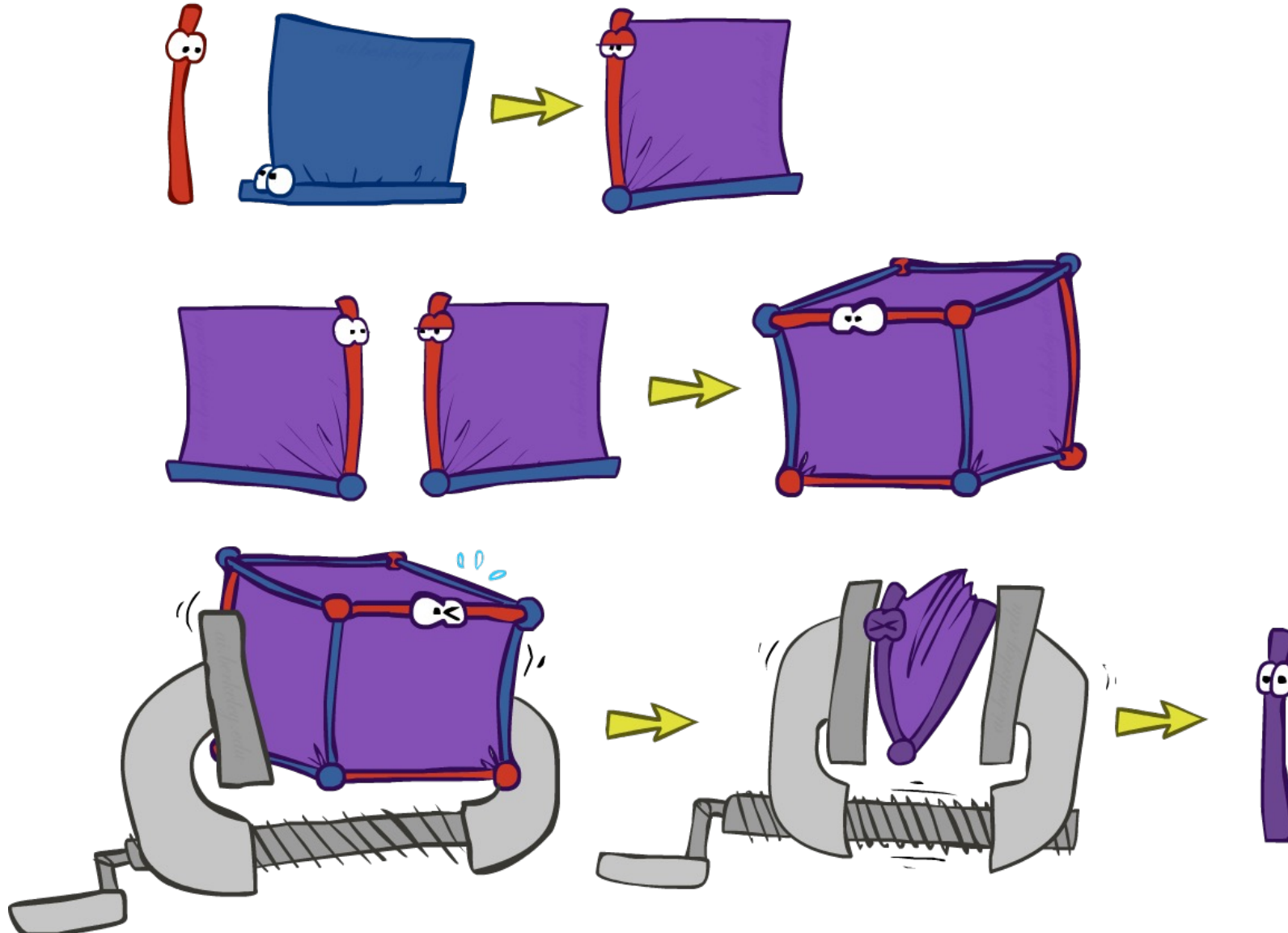
→

$P(L)$

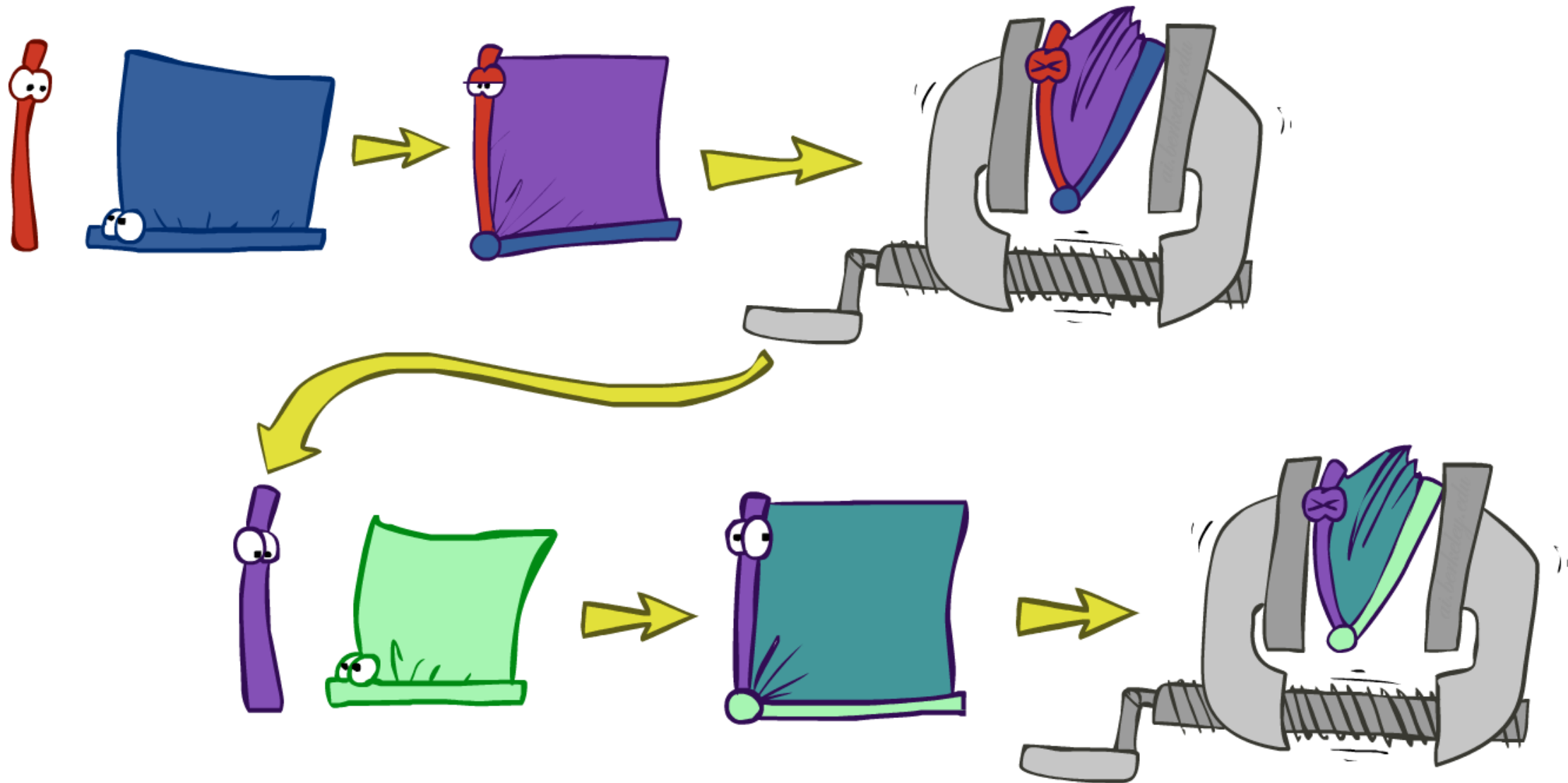
L	
+l	0.134
-l	0.886



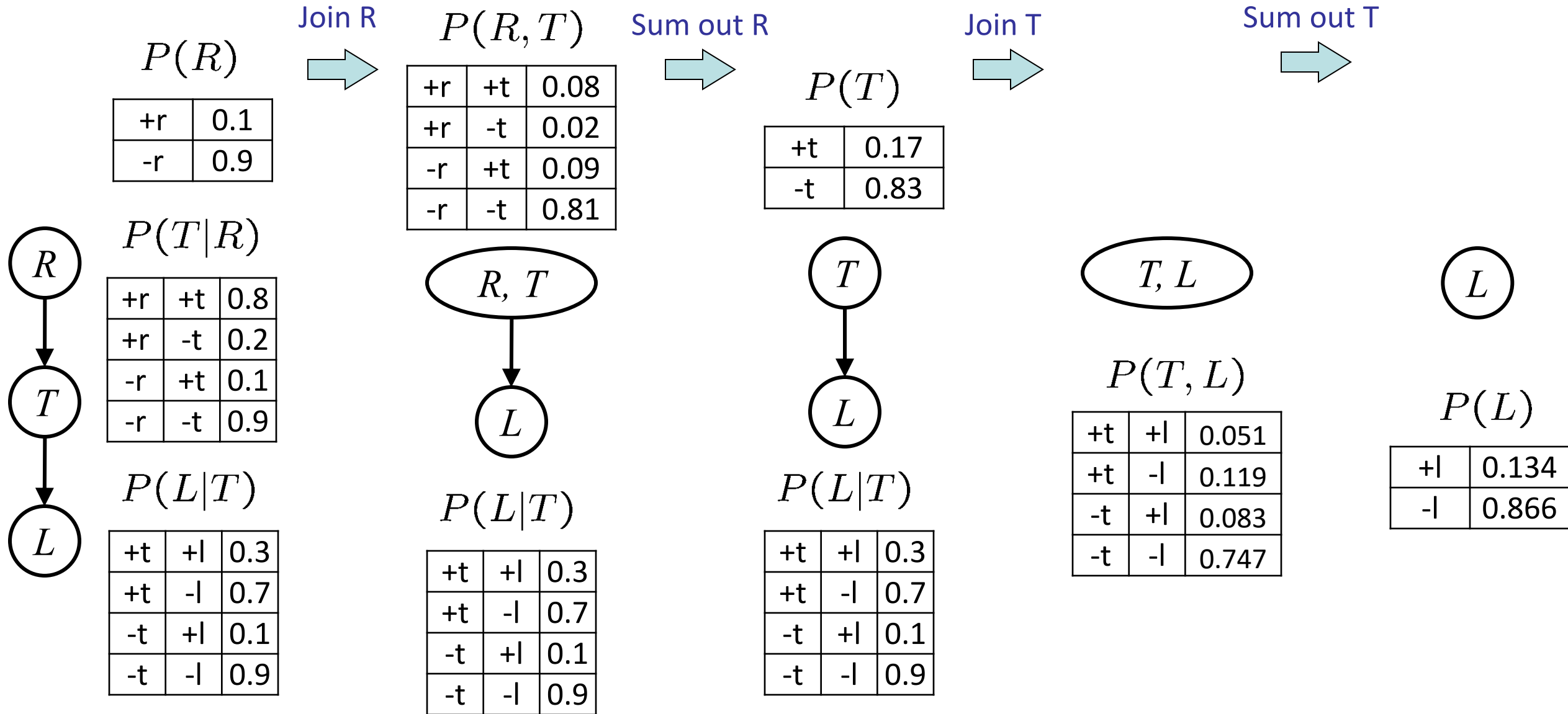
Thus Far: Multiple Join, Multiple Eliminate (= Inference by Enumeration)



Marginalizing Early (= Variable Elimination)



Marginalizing Early! (aka VE)



Evidence

- If evidence, start with factors that select that evidence

- No evidence uses these initial factors:

$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- Computing $P(L|+r)$ the initial factors become:

$$P(+r)$$

+r	0.1
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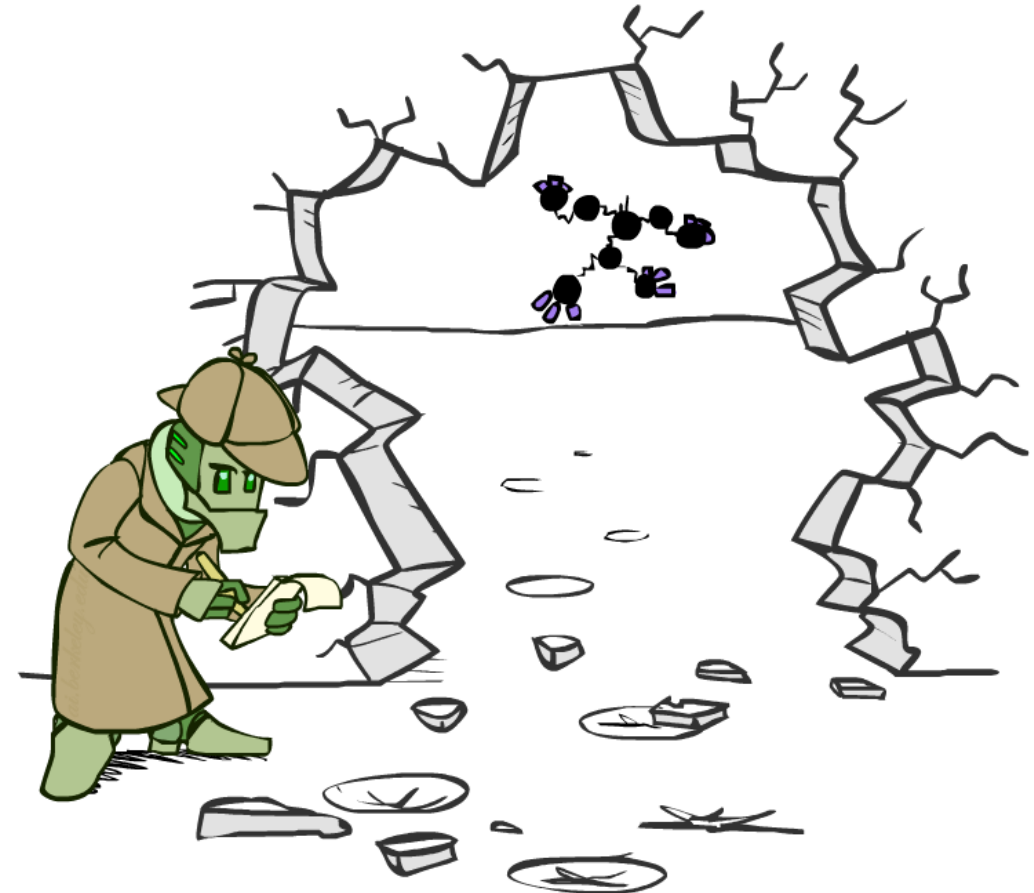
$$P(T|+r)$$

+r	+t	0.8
+r	-t	0.2

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

- We eliminate all vars other than query + evidence



Evidence II

- Result will be a selected joint of query and evidence
 - E.g. for $P(L \mid +r)$, we would end up with:

$$P(+r, L)$$

+r	+l	0.026
+r	-l	0.074

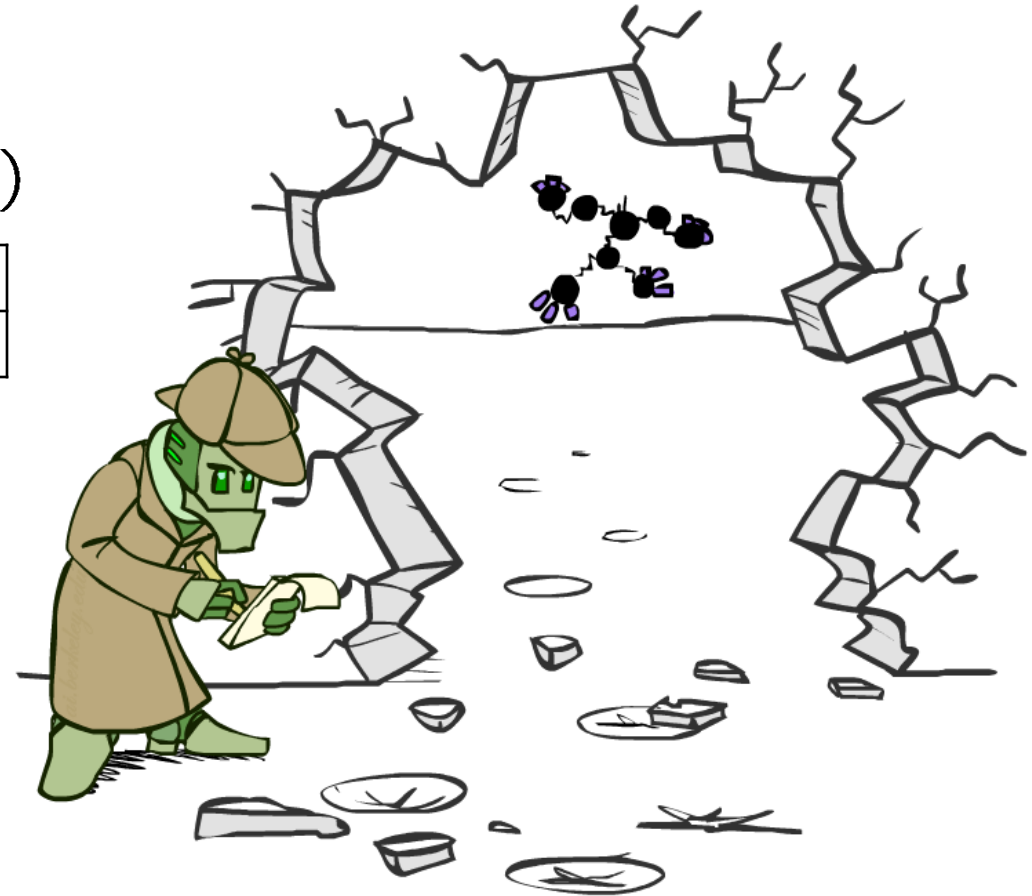
Normalize



$$P(L \mid +r)$$

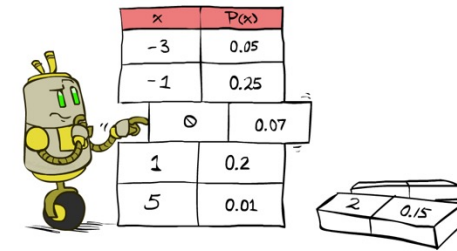
+l	0.26
-l	0.74

- To get our answer, just normalize this!
- That's it!

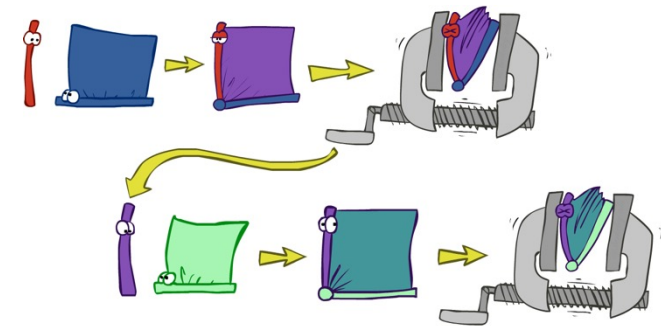


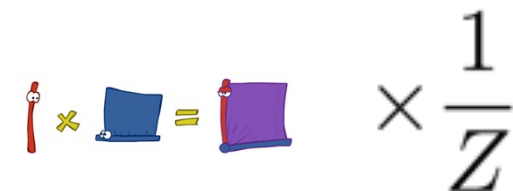
General Variable Elimination

- Query: $P(Q|E_1 = e_1, \dots, E_k = e_k)$
- Start with initial factors:
 - Local CPTs (but instantiated by evidence)
- While there are still hidden variables (not Q or evidence):
 - Pick a hidden variable H
 - Join all factors mentioning H
 - Eliminate (sum out) H
- Join all remaining factors and normalize



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

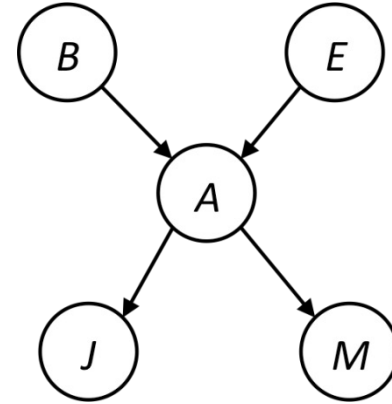



$$\text{red stick} \times \text{blue square} = \text{purple square} \times \frac{1}{Z}$$

Example

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
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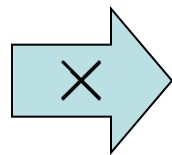


Choose A

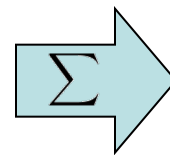
$$P(A|B, E)$$

$$P(j|A)$$

$$P(m|A)$$



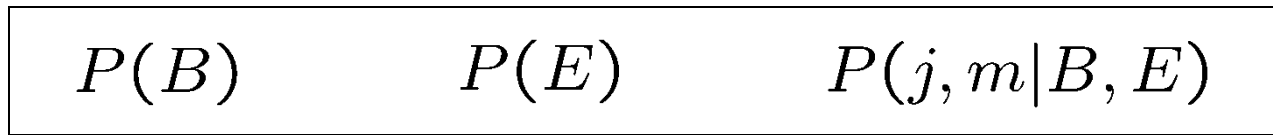
$$P(j, m, A|B, E)$$



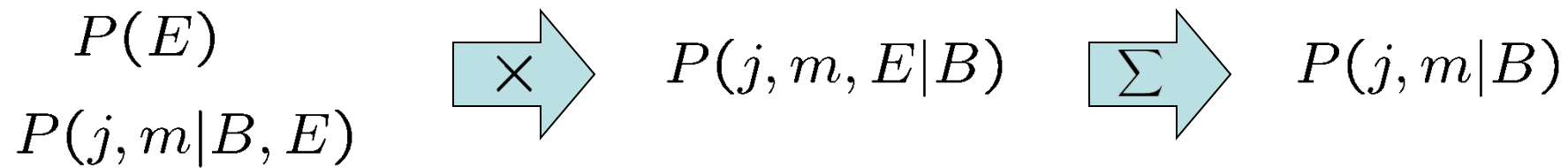
$$P(j, m|B, E)$$

$P(B)$	$P(E)$	$P(j, m B, E)$
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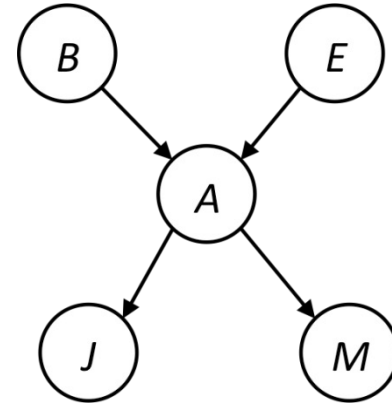
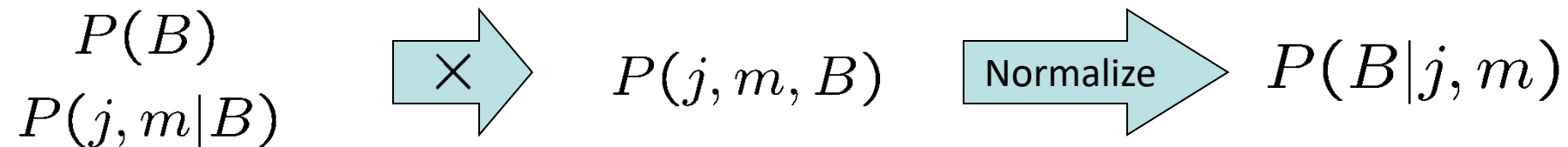
Example



Choose E



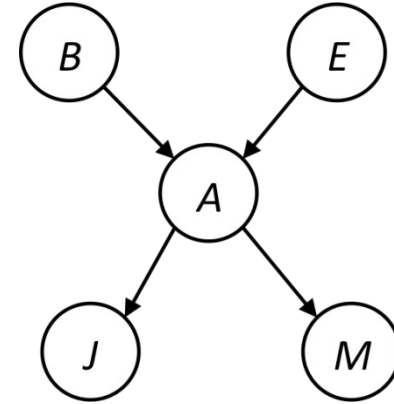
Finish with B



Same Example in Equations

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
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$$\begin{aligned}
 P(B|j, m) &\propto P(B, j, m) \\
 &= \sum_{e, a} P(B, j, m, e, a) \\
 &= \sum_{e, a} P(B)P(e)P(a|B, e)P(j|a)P(m|a) \\
 &= \sum_e P(B)P(e) \sum_a P(a|B, e)P(j|a)P(m|a) \\
 &= \sum_e P(B)P(e)f_1(B, e, j, m) \\
 &= P(B) \sum_e P(e)f_1(B, e, j, m) \\
 &= P(B)f_2(B, j, m)
 \end{aligned}$$

marginal obtained from joint by summing out

use Bayes' net joint distribution expression

use $x^*(y+z) = xy + xz$

joining on a, and then summing out gives f_1

use $x^*(y+z) = xy + xz$

joining on e, and then summing out gives f_2

All we are doing is exploiting $uwv + uwz + uxy + uxz + vwy + vwz + vxy + vxz = (u+v)(w+x)(y+z)$ to improve computational efficiency!

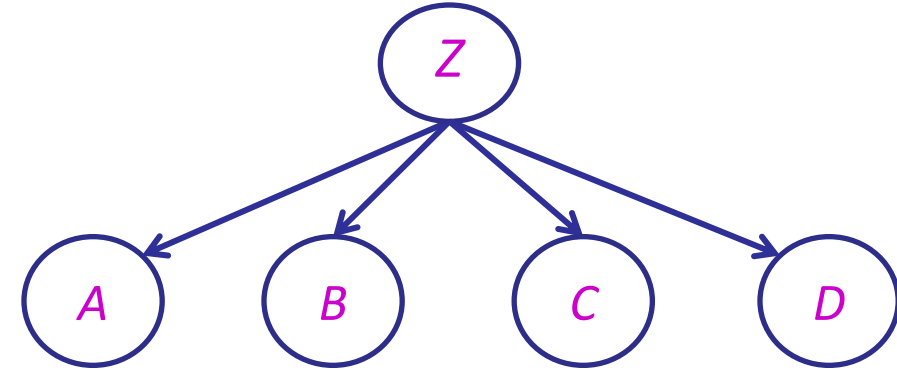
Order matters

- Order the terms Z, A, B C, D

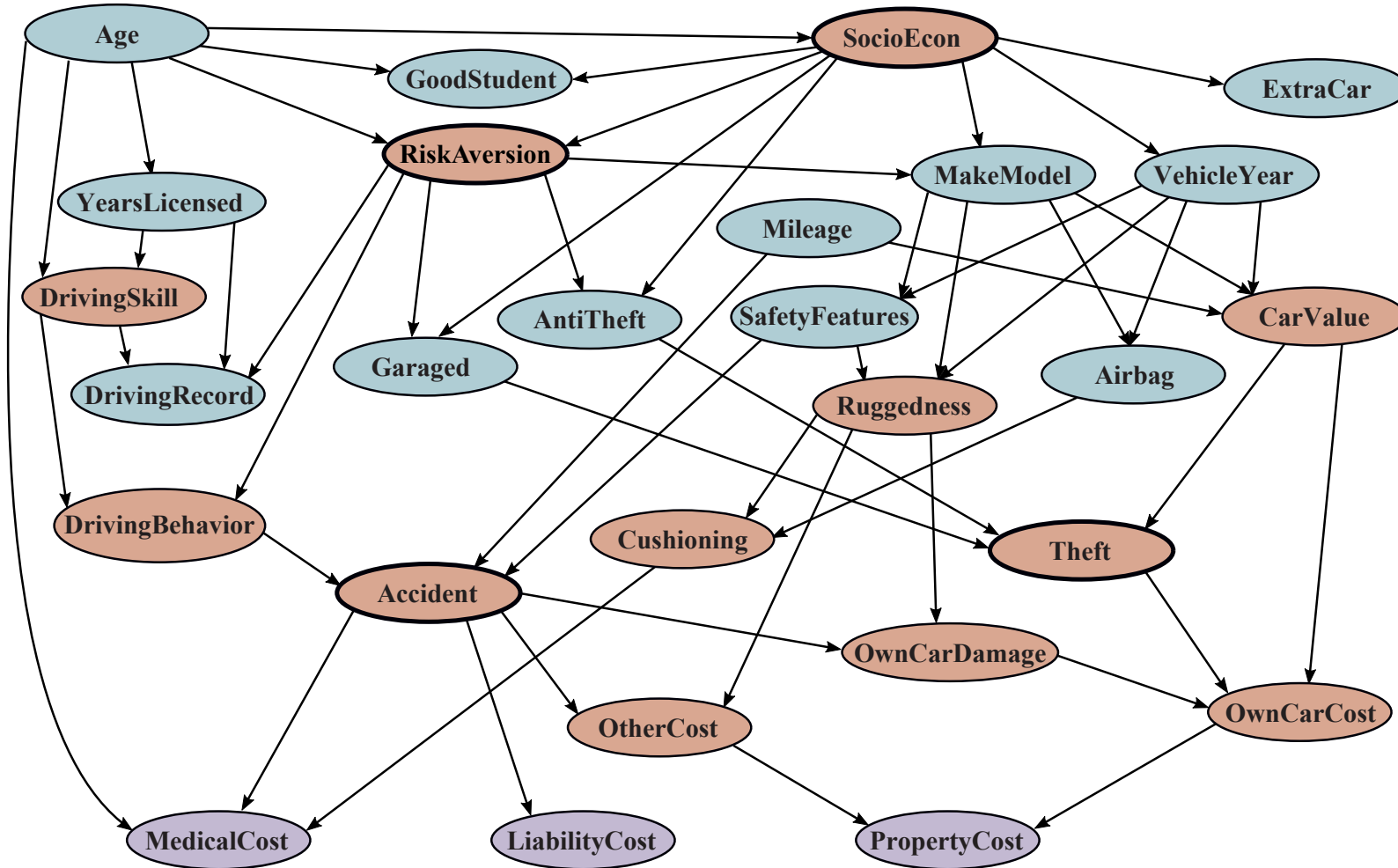
- $P(D) = \alpha \sum_{z,a,b,c} P(z) P(a|z) P(b|z) P(c|z) P(D|z)$
- $= \alpha \sum_z P(z) \sum_a P(a|z) \sum_b P(b|z) \sum_c P(c|z) P(D|z)$
- Largest factor has 2 variables (D,Z)

- Order the terms A, B C, D, Z

- $P(D) = \alpha \sum_{a,b,c,z} P(a|z) P(b|z) P(c|z) P(D|z) P(z)$
- $= \alpha \sum_a \sum_b \sum_c \sum_z P(a|z) P(b|z) P(c|z) P(D|z) P(z)$
- Largest factor has 4 variables (A,B,C,D)
- In general, with n leaves, factor of size 2^n



Example Bayes Net: Car Insurance



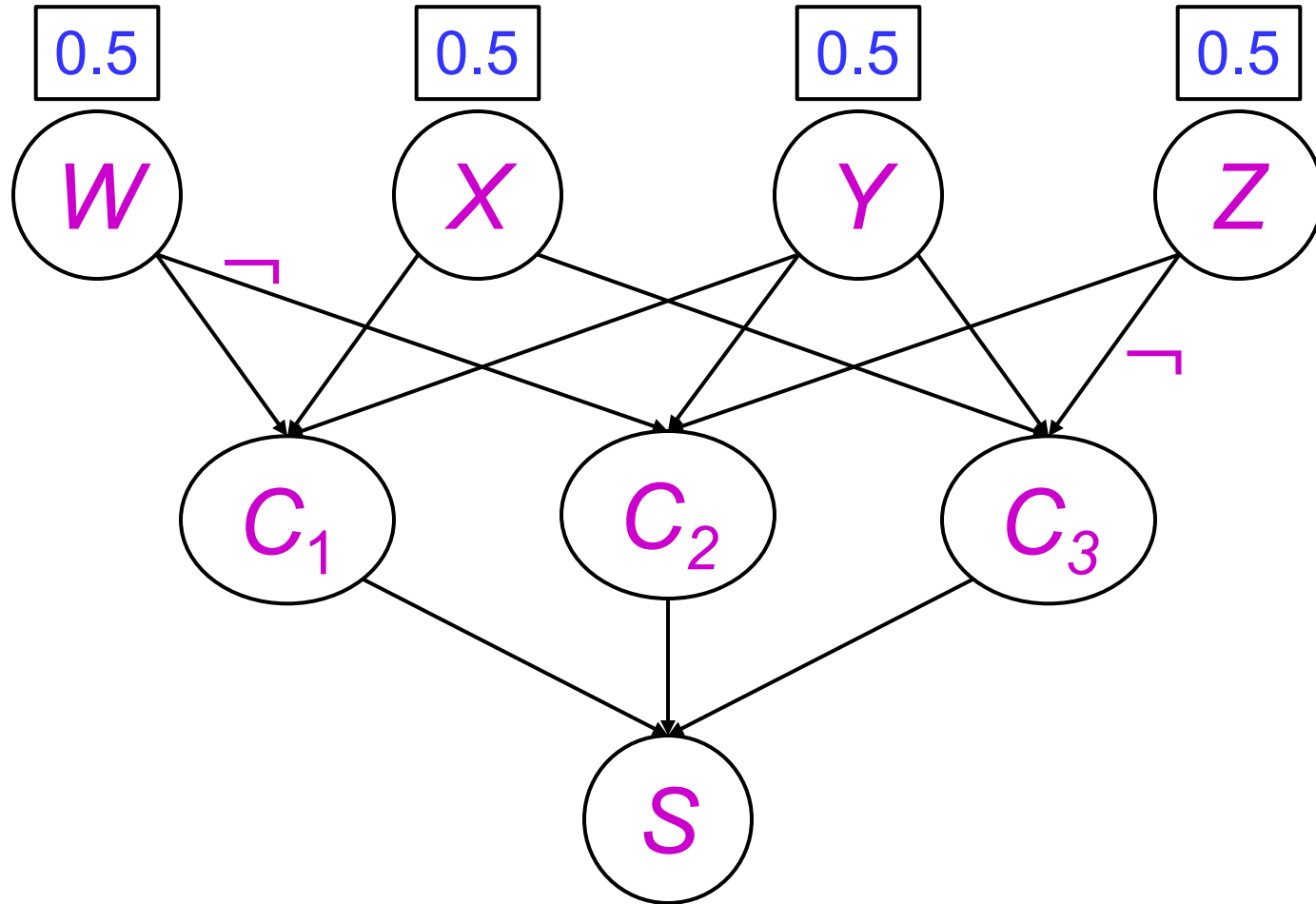
Enumeration: **227M** operations

Elimination: **221K** operations

Computational and Space Complexity

- The computational and space complexity of variable elimination is determined by the largest factor (and it's space that kills you)
- The elimination ordering can greatly affect the size of the largest factor.
 - E.g., ZABCD example 2^n vs. 2
- Does there always exist an ordering that only results in small factors?
 - No!

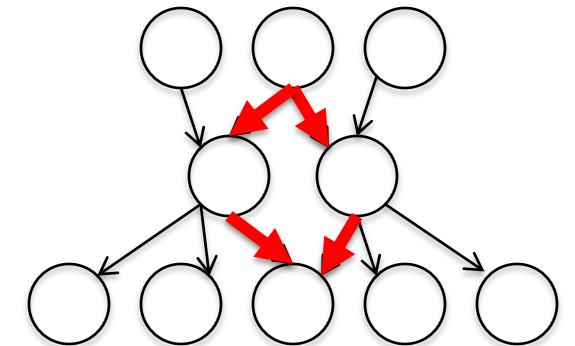
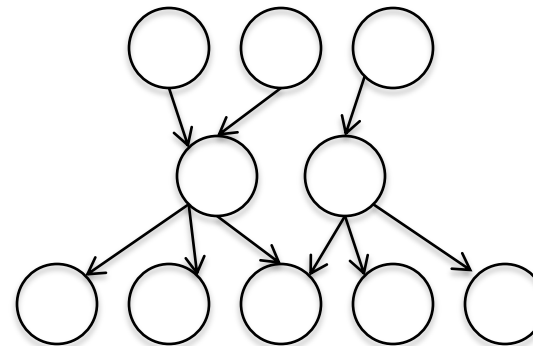
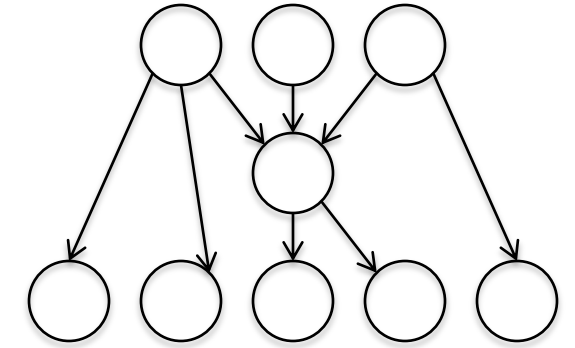
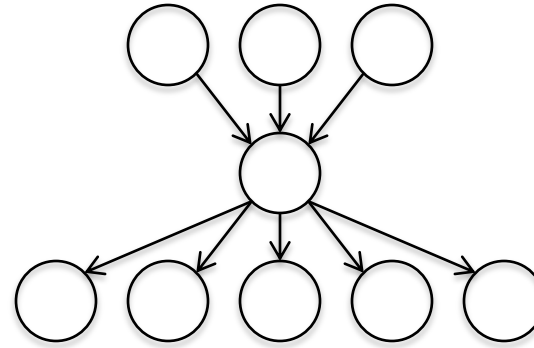
Worst Case Complexity? Reduction from SAT



- Variables: W, X, Y, Z
- CNF clauses:
 1. $C_1 = W \vee X \vee Y$
 2. $C_2 = Y \vee Z \vee \neg W$
 3. $C_3 = X \vee Y \vee \neg Z$
- Sentence $S = C_1 \wedge C_2 \wedge C_3$
- $P(S) > 0$ iff S is satisfiable
 - $\Rightarrow$ **NP-hard**
- $P(S) = K \times 0.5^n$ where K is the number of satisfying assignments for clauses
 - $\Rightarrow$ **#P-hard**

Polytrees

- A polytree is a directed graph with no undirected cycles
- For poly-trees the complexity of variable elimination is **linear in the network size** if you eliminate from the leaves towards the roots
- Cut-set conditioning for near-polytrees
 - Choose set of variables such that if removed, only a polytree remains
 - Solve each polytree separately



Summary

- Exact inference = sums of products of conditional probabilities from the network
- Enumeration is always exponential
- Variable elimination reduces this by avoiding the recomputation of repeated subexpressions
 - Massive speedups in practice
 - Linear time for polytrees
- Exact inference is #P-hard
- Next: approximate inference

